Hangzhou traffic Datasets

Introduction

These datasets provide both the **road network (roadnet.json)** and **traffic flow (flow.json)** data for each scenario, compatible with the CityFlow simulator. They are designed for testing traffic signal control strategies under various real-world and synthetic conditions.

Dataset Information

Dataset 1

Name: hangzhou_1x1_bc-tyc_18041607_1h
Intersection Location: bc-tyc area, Hangzhou, China
Number of intersections: 1
Time span: 3600 seconds (1 hour)
Traffic intensity: Higher traffic congestion
Date in dataset name: 2018-04-16
Dataset type: Real-world traffic data derived from camera observations
Simulator environment: CityFlow
Referred result: 221.03 seconds average travel time
Referred method: SOTL

Dataset 2

Name: hangzhou_1x1_kn-hz_18041608_1h
Location: kn-hz area, Hangzhou, China
Number of intersections: 1
Time span: 3600 seconds (1 hour)
Traffic intensity: Lower traffic congestion
Date: 2018-04-16
Dataset type: Real-world traffic data derived from camera observations
Simulator environment: CityFlow
Referred result: 64.10 seconds average travel time
Referred method: SOTL

Dataset Overview

These datasets are part of a benchmark for **Traffic Signal Control research**. They are designed to evaluate and compare different traffic light control algorithms in realistic urban environments.

Each dataset represents a **traffic scenario at a single intersection** in Hangzhou city, China. The data was collected using traffic cameras that monitor vehicle movements at intersections. The vehicle trajectories and traffic volumes are extracted and then converted into simulation-ready traffic flow files.

The datasets are commonly used with the **CityFlow microscopic traffic simulator**, where each vehicle is simulated individually with its own speed, acceleration, and route.

Dataset Description

he datasets are based on **real camera observations in Hangzhou city**.

However, the camera data did not include accurate records of turning movements. Because of this limitation, the dataset creators defined fixed turning ratios:

- **10% of vehicles turn left**
- **60% of vehicles go straight**
- **30% of vehicles turn right**

Right-turn vehicles are **removed from the traffic signal control problem** because right turns usually do not depend on traffic lights in many traffic systems.

Each road has two lanes:

- **One left-turn lane**
- **One straight lane**

Therefore, each intersection contains **four directions (North, South, East, West)** with two incoming lanes per direction.

Dataset Structure

Each dataset folder contains the following files:

1. roadnet.json

This file describes the **road network structure** used in the simulation.

CityFlow models a road network as a **graph**:

- **Nodes → Intersections**
- **Edges → Roads**

The road network consists mainly of:

- Intersections
- Roads
- road links
- lane links

Essentially, this file defines the **geometry and topology of the intersection**.

2. flow.json

This file describes **vehicle traffic flows**.

It contains information such as:

Vehicle properties:

- vehicle length
- width
- maximum speed
- acceleration
- deceleration

Route information:

- start road
- destination road
- path through the intersection

Traffic generation:

- start time of vehicles
- end time

- interval between vehicles

These parameters define how vehicles **enter the network and move through the intersection**.

Dataset Inputs and Outputs

Input Data

The dataset provides information used by the traffic simulator as input:

- Road network structure
- Lane configuration
- Traffic signal phases
- Vehicle arrival times
- Vehicle routes
- Traffic volume

Typical state information used by traffic control algorithms:

- number of vehicles per lane
- queue length
- vehicle waiting time
- lane occupancy

Output Data

After running a traffic control algorithm, the system measures performance metrics such as:

- **Average travel time**
- **Average waiting time**
- **Queue length**
- **Traffic throughput**
- **Vehicle delay**

The benchmark table reports **Average Travel Time in seconds**, which measures the efficiency of traffic signal control.

Referred Method: SOTL

SOTL = Self-Organizing Traffic Lights

It is a rule-based traffic signal control algorithm.

Main idea:

- traffic lights change depending on traffic demand
- when vehicle queues increase, the signal phase switches

It is often used as a **baseline method** to compare reinforcement learning algorithms.

Applications

This dataset is widely used in research related to:

1. Reinforcement Learning Traffic Control

Training AI agents to control traffic lights.

Example methods:

- DQN
- Multi-agent RL
- Graph neural networks for traffic control.

---

2. Smart City Systems

Developing intelligent traffic management platforms.

---

3. Traffic Simulation Research

Testing new traffic optimization algorithms.

---

4. Autonomous Transportation Research

Studying traffic flow dynamics.

Research Papers

Examples of well-known research:

1. CoLight -2019

CoLight: Learning Network-Level Cooperation for Traffic Signal Control

This paper proposes a **graph attention network (GAT)** to allow multiple traffic signals to cooperate with each other. The method learns the relationships between intersections and improves coordination across the traffic network.

---

2. PressLight –2019 ≈ 600+ citations

PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network

This paper integrates **deep reinforcement learning with max-pressure theory** to optimize traffic signal timing and reduce congestion.

---

3. Advanced-MPLight Expression Might Be Enough: Representing Pressure and Demand for Reinforcement Learning Based Traffic Signal Control

This work improves traffic signal control by introducing a better **traffic state representation** for reinforcement learning algorithms. It proposes **Advanced-MPLight**, which enhances the original MPLight algorithm using improved traffic pressure features.

4. CityFlow Simulator Paper -2019

A Multi-Agent Reinforcement Learning Environment for Large Scale City Traffic Scenario

Efficient traffic simulation platform for RL-based traffic control.